stmindex package description

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This is an interface to facilitate the creation of a single index for `stmlatex`-based documents. It is build upon the imakeidx package.

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1. About

Using `stmindex` it is possible to create a single index for a document very conveniently. When you want to add items to the default index, it is not necessary to specify the index name each time. This is also possible with the basic version of `\index`, however, this makes it easier in case you at some point need another index.

It is assumed that for a large part of documents a single index is sufficient. However, `stmlatex` can also be used as a blueprint to create other indices.

2. Usage

2.1. Preamble

2.1.1. Load the package

Load the package by adding

```
\usepackage[<options>]{stmindex}
```

to your document preamble. For a list of options, see subsubsection 2.1.2.

2.1.2. Options

Option *standard* This is a boolean option. Expected values are either `true` or `false`. It controls whether to load the standard index.

```
\usepackage[standard=true|false]{stmindex}
```

`standard=true` is the default. It is used in case `standard=false` is not set explicitly.

Option *name* This is a string option. It controls the name of the standard index files. It only has an affect in case option `standard=true`.

```
\usepackage[standardname=Testindex]{stmindex}
```

The default is *Index*.

Option *title* This is a string option. It controls the title of the standard index in the document. It only has an affect in case option `standard=true`.

```
\usepackage[standardtitle={Test\ index}]{stmindex}
```

The default is *Index*. Spaces have to be put by spacing commands explicitly.

Option *intoc* This is a boolean option. Expected values are either `true` or `false`. It controls whether to add the standard index to the table of contents. It only has an affect in case option `standard=true`.

```
\usepackage[intoc=true|false]{stmindex}
```

`intoc=true` is the default. It is used in case `intoc=false` is not set explicitly.

Option *columns* This is a integer option. It controls the number of columns used in the index. It only has an affect in case option `standard=true`.

```
\usepackage[columns=3]{stmindex}
```

The default is *3*.

2.1.3. Indexsetup

Use the `\indexsetup` command from `imakeidx` in the preamble to tune the behavior of your index, e.g. the level, e.g.

```
\indexsetup{%
  level=\section*,%
  toclevel=section,%
}
```

in a class with where `\chapter` is the highest level, but you want the index as part of an appendix chapter as a section.

2.2. In the document

`stmindex` introduces a convenient shortcut command to print the standard index in case the option `standard=true`. Use `\printstmindex` at the location of your choosing inside the document to print the standard index.

In case multiple indices are used, the standard `imakeidx` commands apply for printing an index.

3. Create an individual index independent of the standard index

It is possible to create new indices next to or in replacement of the standard index. It is required to load `stmtikzexternalization` after the `\makeindex` call.

Listing 1: Setup individual index

```
% Disable the standard index
\usepackage[
  standard=false
]{stmindex}

% Setup identifiers
\newcommand{\pdKeywordIdxName}{pdkeywordindex}
\newcommand{\pdKeywordIdxTitle}{Peridigm keyword index}

% Macro expansion
\DeclareDocumentCommand{\pdkeywordindex}{%
  O{\pdKeywordIdxName}%style - optional, default in brackets
  m%
}{%
  \index[#1]{#2}%
}

% Make index
\makeindex[%pdkeywords
  name={\pdKeywordIdxName},%
  title={\pdKeywordIdxTitle},%
  intoc,%
  columns=2,%
]

% Print command
\newcommand{\printpdkeywordindex}{\printindex[\pdKeywordIdxName]}

% Must come after \makeindex from imakeidx package, e.g. in stmindex
\usepackage{stmtikzexternalization}
```

4. Known issues

There seems to be a problem with the combination of `imakeidx` and the externalization of `tikz` pictures as from `stmtikz`. Externalization must be activated after the call to `imakeidx` to make it work. See https://tex.stackexchange.com/q/393697.

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A. The code

A.1. stmindex.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                             %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This file includes an interface to
% indices for a LaTeX document
% Based upon the imakeidx package:
%   https://ctan.org/pkg/imakeidx
%
% Usage
%  - Preamble:
%    - \usepackage{imakeidx}
%    - \makeglossaries
%  - Document: e.g. (Adapt to your type of glossary item)
%    - \printindex[type=\acronymtype] or
%    - \printglossary[type=\acronymtype,nonumberlist]
%  - Compilation: e.g. (Adapt to your type of glossary item)
%    - makeindex -s [MYTEXFILENAME].ist -o [MYTEXFILENAME].acr [MYTEXFILENAME].acn
%
% Revisions: 2019-10-27 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
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%                                 __/|__
%                                /_/_/_/
%            www.dlr.de/sy/en      |/ DLR
%
% Copyright (C) 2019-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                            %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

```

```
% Provide the name of your page, the date it was last updated, and a
    comment about what it's used for
\ProvidesPackage{stmindex}[2023/02/12 STMs custom LaTeX index definitions
    ]

% ----------------------------
% Options
% ----------------------------

% For options
\@ifpackageloaded{kvoptions}{}{\RequirePackage{kvoptions}}%

% Option group
\SetupKeyvalOptions{%
  family=stmindex,%
  prefix=stmindex@,%
  setkeys=\kvsetkeys,%
}

% Standard index
\DeclareBoolOption[true]{standard}

% Acronyms
\DeclareStringOption[Index]{standardname}

% Acronyms
\DeclareStringOption[Index]{standardtitle}

% Load styles
\DeclareBoolOption[true]{styles}

% Apply style name
\DeclareStringOption[stmdefault]{style}

% Put index in toc
\DeclareBoolOption[true]{intoc}

% No new page
%\DeclareBoolOption[false]{nonewpage}

% Number of columns
\DeclareStringOption[2]{columns}

% Process options
```

```
\ProcessKeyvalOptions{stmindex}

% ----------------------------
% Modules
% ----------------------------

% Load the base package
\@ifpackageloaded{stmindexbase}{}{%
  \RequirePackage{stmindexbase}%
}%

% Load the styles
\ifstmindex@styles
  \@ifpackageloaded{stmindexstyles}{}{%
    \RequirePackage[
      style=\stmindex@style
    ]{stmindexstyles}%
  }%
\fi

% Load the standard index
\ifstmindex@standard
  \@ifpackageloaded{stmindexstandard}{}{%
    \RequirePackage[%
      name=\stmindex@standardname,%
      title=\stmindex@standardtitle,%
      intoc\ifstmindex@intoc\else{=false}\fi,%
      columns=\stmindex@columns,%
    ]{stmindexstandard}%
  }%
\fi

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                          %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
    this file. LaTeX will ignore anything after this line.
\endinput
```

A.2. stmindexbase.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                             %
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This file includes an interface to
% indices for a LaTeX document
% Based upon the imakeidx package:
%   https://ctan.org/pkg/imakeidx
%
% Usage
%  - Preamble:
%    - \usepackage{imakeidx}
%    - \makeglossaries
%  - Document: e.g. (Adapt to your type of glossary item)
%    - \printindex[type=\acronymtype] or
%    - \printglossary[type=\acronymtype,nonumberlist]
%  - Compilation: e.g. (Adapt to your type of glossary item)
%    - makeindex -s [MYTEXFILENAME].ist -o [MYTEXFILENAME].acr [MYTEXFILENAME].acn
%
% Revisions: 2019-10-27 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
%
%                                  __/|__
%                                 /_/_/_/
%            www.dlr.de/sy/en       |/ DLR
%
% Copyright (C) 2019-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                          %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a comment about what it's used for
\ProvidesPackage{stmindexbase}[2023/02/12 STMs custom LaTeX index basics]

% For options
%\@ifpackageloaded{kvoptions}{}{\RequirePackage{kvoptions}}%

```

```
% ---------------------------
% Options
% ---------------------------

\SetupKeyvalOptions{%
  family=stmindexbase,%
  prefix=stmindexbase@,%
  setkeys=\kvsetkeys,%
}

% Put index in toc
%\DeclareBoolOption[false]{nonewpage}

% Process options
%\ProcessKeyvalOptions{stmindexbase}

% ---------------------------
% Base
% ---------------------------

% If not loaded in advance, load the glossaries package with some default
    options
\@ifpackageloaded{imakeidx}{}{%
  \RequirePackage%[%
    %\ifstmindexbase@nonewpage nonewpage\fi
  %]
  {imakeidx}%
}%

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                        %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
   this file. LaTeX will ignore anything after this line.
\endinput
```

A.3. stmindexstandard.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                           %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This file includes an interface to
```

```
% indices for a LaTeX document
% Based upon the imakeidx package:
%   https://ctan.org/pkg/imakeidx
%
% Revisions: 2019-10-27 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
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%                                 __/|__
%                                /_/_/_/
%            www.dlr.de/sy/en      |/ DLR
%
% Copyright (C) 2019-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                            %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a
    comment about what it's used for
\ProvidesPackage{stmindexstandard}[2023/02/12 STMs custom LaTeX standard
    index definitions]

% For new commands
\@ifpackageloaded{xparse}{}{\RequirePackage{xparse}}%
\@ifpackageloaded{ifmtarg}{}{\RequirePackage{ifmtarg}}%

% ----------------------------
% Options
% ----------------------------

% For options
\@ifpackageloaded{kvoptions}{}{\RequirePackage{kvoptions}}%

% Option group
\SetupKeyvalOptions{%
  family=stmindexstandard,%
  prefix=stmindexstandard@,%
  setkeys=\kvsetkeys,%
```

```
}

% Name
\DeclareStringOption[Keywords]{name}

% Title
\DeclareStringOption[Keywords]{title}

% Put index in toc
\DeclareBoolOption[true]{intoc}

% No new page for index
%\DeclareBoolOption[false]{nonewpage}

% Number of columns
\DeclareStringOption[2]{columns}

% Put index in toc
%\DeclareBoolOption[true]{columnseprule}

% Process options
\ProcessKeyvalOptions{stmindexstandard}

% ----------------------------
% Base
% ----------------------------

% If not loaded in advance, load the base package with some default
    options
\@ifpackageloaded{stmindexbase}{}{%
  \RequirePackage{stmindexbase}%
}%

% ----------------------------
% Index name keywords
% ----------------------------

% Keyword index name
\newcommand{\stmStandardIdxName}{\stmindexstandard@name}

% ----------------------------
% Commands
% ----------------------------

```

```
% Expansion of macros by identity function
% https://tex.stackexchange.com/a/399451/44634
% https://tex.stackexchange.com/a/33753
\DeclareDocumentCommand{\stmindex}{%
  O{\stmStandardIdxName}%style - optional, default in brackets
  m% date/year
}{%
  \index[#1]{#2}
}

\makeindex[%
  name={\stmindexstandard@name},%
  title={\stmindexstandard@title},%
  intoc\ifstmindexstandard@intoc\else{=false}\fi,
  columns=\stmindexstandard@columns,%
]

\newcommand{\printstmindex}{\printindex[\stmStandardIdxName]}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                        %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
    this file. LaTeX will ignore anything after this line.
\endinput
```

A.4. stmindexstyles.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                           %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This file includes index setup
% Based upon the index package:
%   https://ctan.org/pkg/index
%
% Usage
%  - Preamble:
%    - \usepackage{stmindexstyles}
%
% Revisions: 2019-10-27 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
%
```

```
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
%
%                                 __/|__
%                                /_/_/_/
%            www.dlr.de/sy/en      |/ DLR
%
% Copyright (C) 2019-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                            %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a
    comment about what it's used for
\ProvidesPackage{stmindexstyles}[2023/02/12 STMs index style definitions]

% ---------------------------
% Options
% ---------------------------

% For options
\@ifpackageloaded{kvoptions}{}{\RequirePackage{kvoptions}}%

\SetupKeyvalOptions{%
  family=stmindexstyles,%
  prefix=stmindexstyles@,%
  setkeys=\kvsetkeys,%
}

% Style name
\DeclareStringOption[stmdefault]{style}

% Process options
\ProcessKeyvalOptions{stmindexstyles}

% ---------------------------
% Style
% ---------------------------

\@namedef{stmindexstyles@style@default}{%
```

```
  % This here is empty to load the standard index setup
}

\@namedef{stmindexstyles@style@stmdefault}{%
  \indexsetup{%
    firstpagestyle=plain%
  }%
}

% Evaluate option settings
% https://tex.stackexchange.com/a/249102
\newcommand*{\stmindexstyles@evaluate}[1]{%
  % Check, if option value in \<prefix>@<option> exists
  \@ifundefined{stmindexstyles@#1}{%
    % Should not happen
    \PackageError{stmindexstyles}{Evaluating unknown option `#1'}\@ehc
  }{%
    % Check, if there is an implementation for the value of the option
    % in macro \<prefix>@<option>@<value>
    \@ifundefined{stmindexstyles@#1@\csname stmindexstyles@#1\endcsname}{
    %
      \PackageError{stmindexstyles}{%
        Unknown option setting: #1=%
        \csname stmindexstyles@#1\endcsname
      }\@ehc
    }{%
      % Call the implementation for the value
      \csname stmindexstyles@#1@%
              \csname stmindexstyles@#1\endcsname
      \endcsname
      \relax
    }%
  }%
}

\stmindexstyles@evaluate{style}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                          %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
    this file. LaTeX will ignore anything after this line.
\endinput
```